
Algorithm 1 Generate Word Search Puzzle

```
1: Function generateGrid(wordList, gridSize)
2:   size  $\leftarrow$  gridSize
3:   Create an empty 2D matrix[size][size]
4:
5:   for each cell in the matrix do
6:     Fill the cell with a placeholder (like '.')
7:
8:   end for
9:
10:  for each word in wordList do
11:    placed  $\leftarrow$  false
12:
13:    while placed = false do
14:      Pick a random direction (horizontal, vertical, or diagonal)
15:      Randomly choose a starting (x, y) position
16:
17:      if the word actually fits in that direction then
18:        placeWord(word, startX, startY, direction)
19:        placed  $\leftarrow$  true
20:
21:      end if
22:
23:    end while
24:
25:  end for
26:
27:  for each remaining empty cell do
28:    Replace placeholder with a random A–Z letter
29:
30:  end for
31:  return matrix
32: End Function
```

Algorithm 2 Place a Word in the Grid

```
1: Function placeWord(word, x, y, direction)
2:
3: for  $i \leftarrow 0$  to length(word) - 1 do
4:
5:   if direction = "horizontal" then
6:     matrix[y][x + i]  $\leftarrow$  word[i]
7:
8:   else if direction = "vertical" then
9:     matrix[y + i][x]  $\leftarrow$  word[i]
10:
11:   else if direction = "diagonal" then
12:     matrix[y + i][x + i]  $\leftarrow$  word[i]
13:
14:   end if
15:
16: end for
17: End Function
```

Algorithm 3 Check User-Selected Word Coordinates

```
1: Function checkWord(startX, startY, endX, endY, wordList, grid)
2:   selectedWord  $\leftarrow$  ""
3:   dx  $\leftarrow$  sign(endX - startX)
4:   dy  $\leftarrow$  sign(endY - startY)
5:   x  $\leftarrow$  startX
6:   y  $\leftarrow$  startY
7:
8:   while (x, y) hasn't reached (endX, endY) do
9:     selectedWord += grid[y][x]
10:    x  $\leftarrow$  x + dx
11:    y  $\leftarrow$  y + dy
12:
13:   end while
14:   selectedWord += grid[endY][endX]
15:
16:   if selectedWord is in wordList then
17:     return true
18:
19:   else
20:     return false
21:
22:   end if
23: End Function
```
